

- **Topic Modelling** is an **unsupervised NLP** technique that automatically identifies **topics or themes** in a large collection of documents, based on the words they contain.

Why ?

1. In today's data-driven world, there's a vast amount of unstructured text (news, reviews, articles, tweets).
2. Topic modelling helps **categorize** these documents automatically.
3. It allows us to **discover hidden structures** or themes in text without any prior labels.

Topic Modelling vs Topic Classification

Feature	Topic Modelling	Topic Classification
Learning Type	Unsupervised	Supervised
Input	Raw text data	Labeled data
Output	Hidden topics (discovered)	Predicted labels
Example	Discovering new topics in news articles	Classifying emails as spam/non-spam

Latent Dirichlet Allocation (LDA)

LDA — *Latent Dirichlet Allocation* — is one of the most widely used algorithms for topic modelling.

Meaning:

Term	Meaning
Latent	Hidden (topics are not directly observed)
Dirichlet	A type of probability distribution assumed for topics and words
Allocation	Assigning topics to documents and words

⚙️ How LDA Works

LDA assumes:

- Each **document** is a **mixture of topics**.
- Each **topic** is a **mixture of words**.
- Documents are generated by randomly choosing topics, then words from those topics.

📋 LDA Parameters

Hyperparameter	Role
α (alpha)	Controls the number of topics per document. (Higher $\alpha \rightarrow$ more mixed topics)
β (beta)	Controls the number of words per topic. (Higher $\beta \rightarrow$ topics are more spread out)
K	The number of topics to extract.

🧩 Example – Step-by-Step LDA Process

Documents

1. Doc 1: *"I like to eat broccoli and bananas."*
2. Doc 2: *"I ate a banana and spinach smoothie."*
3. Doc 3: *"Cherries are my favorite fruit."*

Vocabulary

V = 16 unique words

[broccoli, bananas, eat, like, to, and, I, ate, a, spinach, smoothie, cherries, are, my, favorite, fruit]

Step 1 : Create Document-Word Matrix

Document	broccoli	bananas	eat	like	...	favorite	fruit
Doc 1	1	1	1	1	...	0	0
Doc 2	0	1	0	0	...	0	0
Doc 3	0	0	0	0	...	1	1

Step 2 : Initialize Topics

Choose $K = 2$ topics, randomly assign words:

Word	Doc 1 Topic	Doc 2 Topic	Doc 3 Topic
broccoli	1	–	–
bananas	1	1	–
...	...	...	...
cherries	–	–	2
favorite	–	–	2

Step 3 : Count Words by Topic

Word	Topic 1 Count	Topic 2 Count
broccoli	1	0
bananas	2	0
cherries	0	1
favorite	0	1

Step 4 : Update Topic Probabilities

For each word, LDA updates its topic based on how frequently it appears with other words in a topic and in a document.

This process repeats iteratively (Gibbs Sampling).

Step 5 : Converge to Final Topics

After several iterations:

Topic	Words
Topic 1	broccoli, bananas, eat, like, spinach, smoothie
Topic 2	cherries, are, my, favorite, fruit

✓ Interpretation:

- Topic 1 → Food/Recipes
- Topic 2 → Fruits/Favorites

